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Education

PhD in Economics, University College London	2021 - Present
MRes in Economics, University College London	2020 - 2021
MSc in Economics, University College London	2019 - 2020
BSc in Economics, University College London	2016 - 2019

References

Wei Cui	Franck Portier	Lukasz Rachel
Professor of Economics	Professor of Economics	Associate Professor
University College London	University College London	University College London
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Research Interests

Macroeconomics, Financial Intermediation, Technological Changes

Working Papers

Bank Market Power and Interest Rate Risk (Job Market Paper)

What determines banks' earnings exposure to interest rate risk? We propose a novel mechanism linking optimal exposure to relative market power in funding and lending. Two stylised facts motivate the analysis: banks partially match income and expense sensitivities but retain heterogeneous exposures to interest rate changes, while loan pricing suggests stronger market power over fixed-rate than variable-rate contracts. In our model, risk-averse banks adjust their loan portfolios to balance earnings stability against exploiting pricing power over fixed-rate lending. Relative market power determines whether earnings rise or fall when rates increase. A calibrated extension shows that stronger deposit competition shifts banks' exposure while leaving fixed-rate lending nearly unchanged. In this calibration, greater interest rate uncertainty raises borrowers' demand for protection, expanding fixed-rate lending and making earnings vulnerable to rate increases. These findings connect competition on both sides of banks' balance sheets to the allocation of interest rate risk, with implications for competition policy and supervision.

Technological Innovations and Worker Skills

With Matthew Nibloe, Peter Nunez and Joern Onken

This paper studies how workers respond to skill-specific technological innovations through investment in training. We construct a novel measure of technological exposure by combining patent texts with O*NET skill descriptors using natural language processing. Linking this measure to UK longitudinal worker data, we find that greater technological exposure increases hourly wages, particularly among highly educated workers and those in non-routine occupations. Exposure also raises participation in training, especially training for future employment opportunities, and increases voluntary employer mobility. We develop a model of multidimensional skill investment under uncertain future skill returns. Higher skill returns provide more resources to pay for training, and allowing training to respond amplifies the output gains from technological change. Greater uncertainty about future skill returns, such as due to the unpredictable labour-market effects of AI, encourages workers to acquire a more balanced set of skills, giving them more flexibility to respond as skill returns change. These findings highlight workers' proactive investment in skills and pursuit of new employment opportunities as important margins of adjustment to technological innovation.

Bank Size and Wholesale Funding Supply

When a depository bank loses net worth, does it contract its own lending or the funding it provides to nonbank lenders? In U.S. Call Reports, mortgage-backed and other asset-backed securities account for roughly 6% of assets for smaller banks and nearly 15% in for largest banks. I develop a model in which a wholesale claim funds a nonbank lender and serves as a private safe asset for the depository bank that holds it. Small banks hold just enough claims to meet liquidity requirements and cut both forms of lending after a loss. In contrast, large banks absorb a marginal loss of balance-sheet capacity entirely by cutting wholesale funding. In a U.S. banking industry calibration, a 50-basis-point decline in annualised loan returns reduces wholesale funding by about 1.6% on impact, compared with just over 0.5% with a representative depository bank, highlighting the role of heterogeneity in credit supply responses.

Work in Progress

Public Debt, Bank Liquidity and Credit Supply

Treasury securities provide banks with liquidity and offer savers an alternative to deposits. How does this dual role shape the effect of public liquidity on bank lending? I construct Treasury issuance surprises from within-quarter errors in the Federal Reserve's Greenbook forecasts of U.S. government net borrowing. Using quarterly bank balance sheets from 1994 to 2019, I estimate lending responses across banks ranked by high-quality liquid assets (HQLA) as a share of total assets. For a 1% increase in government debt, real lending in the most liquid quintile rises by about 1.4%. In contrast, real lending contracts for banks with less HQLA. These preliminary findings motivate a model of how competition for deposits and liquidity requirements shape credit supply.

Presentations

2025 HEC Paris PhD Conference, Institute for Fiscal Studies* (IFS), Organisation for Economic Co-operation and Development* (OECD)
* Joint work presented by a co-author.

Teaching Experience

ECON0107: MRes Macroeconomics (UCL) 2022-2026
1st-Year PhD – Teaching Assistant to Franck Portier and Lukasz Rachel

ECON0016: BSc Macroeconomics (UCL) 2021-2026
2nd-Year BSc – Teaching Assistant to Wei Cui, Wendy Carlin, Marta Morazzoni and Franck Portier

Awards and Scholarships

UCL Economics Departmental Award: Best MRes TA 2024

UCL Studentship 2021-Present

Professional Experience

University College London – Research Assistant to Wei Cui 2023-Present
Time series analysis of UK house price and mortgage issuance response to monetary and fiscal shocks.

Bank of England – Intern, Stress Testing Strategy Division 2018
Analysis of short-term funding data of systematically important banks as input to stress testing design.

Technical and Language Skills

Technical: Python, MATLAB, Stata.
Languages: English (fluent), Chinese (native).